

Grey Water

Grey water is the waster water generated from laundry, bathing, dish washing comprises 50-80 percent of all household waste water

Another interesting aspect about the society is that none of the houses have a geyser to heat water. Thanks again to Chandra who set up a solar water heater on the terrace. "Geysers consume a lot of power, so we thought of installing a solar water heater on our terrace, which will provide hot water to the residents," he said.

Chandra showed us the solar water heating panels which were placed one level above the terrace. Placed in a 4 by 4 grid like arrangement called batch heaters, the water to the panels is provided by a pipe that runs across all the panels. It's technically called an Integrated Collector Storage tank. Each heater has a glass covering and is placed at an angle from the ground. It's fed water from one end. The water trickles down through the miniature capillaries such that by the time it gets out of one batch of four heaters, it is comparatively warm. The process is repeated for three more such batches.

"Sometimes during the summers, with soaring temperatures to prevent overheating, I have to let out some steam from these steam relievers," demonstrated Chandra. After the water has passed through all the panels, it is considerably hot and is collected in a 2000-litre tank which regulates the temperature. When we were checking it out, the temperature of the water was 60 degrees Celcius. "This heater is connected to the all the 25 bathrooms in the building, thus bypassing the need for a geyser," said Chandra. Just doing away with geysers cuts down on 30 units of power per month which equates to Rs. 150 (at a rate of Rs.5/unit) savings.

Hybrid Energy: Solar + Wind

The next logical question on our mind was harnessing Wind energy. But after talking to an energy technology specialist, we learned that using just windmills to generate

energy from wind is just not a viable option.

Sunil Tongay, an electronics graduate from Mumbai University had always been interested in electronics and indulged in a lot of hobby electronics in his school days. After working for around 15 years in the IT industry, Tongay decided to move into the Energy Technology sector back in 2005. He felt that the future lay in Energy as IT was reaching a saturation point.

"I am deeply interested in electronics. I decided to take a break from IT because I thought that my base in electronics will help me a lot in the energy technology segment," he told us recalling his decision. Five years into the field Tongay has spearheaded around 100 projects which involved using alternate sources of energy like wind and solar energy in daily life. He specialises in hybrid energy generation.

"The minimum wind speed required to generate electricity is 5m/sec. Now in a city like Mumbai where most of the housing societies are not planned correctly (ie houses facing the south) one cannot efficiently harness the wind energy. Housing societies in Worli

have an edge over other places in Mumbai as the wind speed there is quite high. And apart from the monsoon seasons, there isn't much wind blowing anyway," said Tongay justifying his stand as to why only using a windmill in societies was not a viable option.

However if windmills are teamed up with solar panels, the combination can work wonders. During the monsoon months, the wind mills can contribute more and during summers the

First Windmill

Charles F. Brush made the first use of a large windmill to generate electricity, in 1888 in Cleveland, Ohio

solar panels can take over. During other months, it will always work in a combined form thus harness the best of both natural sources.

Tongay has installed a hybrid system for his office premises. He took us to his terrace where he showed us an interesting aspect which involved using alternative sources as well as energy from the grid to reduce load on the grid.

"We have teamed the hybrid solar and wind system to a Grid Tied Inverter (GTI). The GTI allows us to combine the electricity generated from the hybrid solar

– wind system along with the grid to power a load irrespective of its energy consumption. We have our server room in the basement, which needs 24/7 air conditioning. Thanks to the GTI we can use the grid power along with the power from hybrid system to power those ACs. Due to this we save a lot of power which we could have otherwise used from the grid," explained Tongay. He also drew up a circuit diagram

which simplified the understanding of the concept. However, this is effective only in places which have a constant power supply from the grid.

Suresh Patkar, a Navi Mumbai resident also installed a hybrid solar-wind energy system at his residence as his family's consumption was quite high for a three member family. "I was planning



Energy consultant Sunil Tongay

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GRID TIE INVERTER

A grid tie inverter acts as a mediator between your DC solar array and your home's AC power grid. The DC power generated by the solar panels or hybrid solar-wind system is inverted to a higher AC voltage so that home appliances can be used. When not in use, the power generated from the alternate sources can be fed back to the grid. Using a GTI bypasses the need to use battery power bank to store energy. But for the grid tie inverters must be in phase with the power grid for you to send back power to the grid.